

LOCARNO MAGADINO AP, Switzerland

WMO: 067620

Lat: 46.166N

Lon: 8.882E

Elev: 198

StdP: 98.97

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -6.1	(c) -4.5	(d) -15.6	(e) 1.0	(f) 4.3	(g) -13.0	(h) 1.3	(i) 4.9	(j) 9.4	(k) 7.7	(l) 8.2	(m) 7.8	(n) 1.1	(o) 90	(p) 0.416

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	30.7	21.8	29.4	21.1	28.2	20.4	23.0	29.1	22.1	27.9	21.3	26.7	3.0	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.0	16.1	25.9	20.2	15.2	24.9	19.4	14.5	24.1	68.9	29.2	65.6	27.9	62.8	26.7	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.4	(b) 6.0	(c) 4.8	(e) -8.1	(f) 32.4	(g) 2.6	(h) 1.4	(i) -10.0	(j) 33.4	(k) -11.5	(l) 34.2	(m) -13.0	(n) 35.0	(o) -14.9	(p) 36.0		
(4) 7.4	(5) 6.0	(6) 4.8	(7) -8.6	(8) 24.4	(9) 2.5	(10) 0.9	(11) -10.4	(12) 25.0	(13) -11.9	(14) 25.6	(15) -13.2	(16) 26.1	(17) -15.1	(18) 26.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.3	2.1	4.0	8.5	12.2	16.4	20.4	22.2	21.5	17.4	12.5	6.9	2.5
	DBStd	7.61	2.64	2.72	3.02	2.94	2.80	2.80	2.38	2.29	2.64	2.87	3.14	2.90
	HDD10.0	832	244	168	65	12	1	0	0	0	0	9	100	233
	HDD18.3	2546	502	401	306	185	73	11	2	3	48	180	342	492
	CDD10.0	1658	1	1	17	78	198	312	378	357	221	88	8	0
	CDD18.3	328	0	0	0	1	12	74	122	101	19	0	0	0
	CDH23.3	2577	0	0	3	16	117	602	1001	721	114	4	0	0
	CDH26.7	641	0	0	0	2	14	153	283	179	11	1	0	0
Wind	WSAvg	1.6	1.3	1.5	1.8	1.9	1.8	1.8	1.8	1.6	1.5	1.3	1.3	1.2
Precipitation	PrecAvg	1726	73	67	81	148	194	180	167	178	190	180	194	74
	PrecMax	2559	244	278	277	400	553	447	325	319	442	576	742	225
	PrecMin	993	0	4	6	13	71	27	31	21	20	10	3	0
	PrecStd	378	62	62	63	97	104	94	80	74	130	122	176	58
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.6	16.4	22.5	25.2	28.5	31.9	32.2	32.3	28.1	23.2	17.0	13.7
		MCWB	6.8	7.8	11.4	15.3	18.6	21.9	22.7	23.0	20.0	16.0	10.3	5.8
	2%	DB	10.5	13.2	19.1	22.9	26.2	30.0	30.7	30.1	26.2	20.8	14.7	10.4
		MCWB	4.4	6.5	10.4	14.0	17.5	21.4	21.8	21.8	18.8	15.2	9.9	6.1
	5%	DB	8.4	11.3	17.2	21.1	24.4	28.5	29.5	28.6	24.5	19.2	13.3	8.7
		MCWB	3.9	5.7	9.6	12.9	16.5	20.6	21.3	20.9	17.8	14.5	9.8	5.6
	10%	DB	6.7	9.6	15.3	19.2	22.8	26.9	28.3	27.2	23.0	17.7	12.0	7.3
		MCWB	3.3	4.8	8.8	12.0	15.5	19.5	20.4	20.2	16.8	13.9	9.5	4.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.0	9.0	12.5	16.4	19.8	23.7	24.2	23.7	21.2	17.7	12.9	9.5
		MCDB	12.3	14.1	19.6	23.4	26.6	30.2	30.1	30.0	26.2	20.8	14.3	10.5
	2%	WB	6.1	7.5	11.5	14.8	18.6	22.3	23.1	22.7	20.1	16.7	11.9	7.7
		MCDB	8.6	11.5	17.1	21.3	24.7	28.4	29.1	28.7	24.4	19.2	13.3	8.7
	5%	WB	4.9	6.5	10.6	13.7	17.6	21.3	22.1	21.9	19.1	15.7	11.0	6.4
		MCDB	7.2	9.9	15.4	19.3	22.9	27.1	27.9	27.3	22.9	17.9	12.3	7.9
	10%	WB	3.9	5.6	9.6	12.8	16.6	20.4	21.4	21.1	18.2	14.8	10.0	5.2
		MCDB	5.9	8.5	14.1	17.5	21.1	25.6	26.8	25.9	21.5	16.8	11.3	7.0
Mean Daily Temperature Range	5% DB	MDBR	8.5	9.8	11.3	10.7	10.2	10.5	11.0	10.5	9.9	8.9	8.1	8.1
		MCDBR	11.3	13.2	15.4	15.3	13.7	12.9	12.6	12.6	12.2	11.4	10.4	10.0
		MCWBR	6.9	7.8	8.2	7.6	6.5	6.0	5.7	5.8	6.1	6.5	6.3	6.3
	5% WB	MCDBR	9.6	11.7	13.4	13.2	11.8	11.8	11.5	11.4	10.3	8.8	6.7	8.2
		MCWBR	6.5	7.4	7.5	6.8	6.0	5.6	5.5	5.5	5.5	5.3	4.7	5.8
Clear-Sky Solar Irradiance	taub	0.319	0.356	0.419	0.489	0.502	0.518	0.502	0.491	0.468	0.430	0.362	0.314	
	taud	2.274	2.178	2.056	1.937	1.983	1.999	2.058	2.097	2.115	2.169	2.289	2.322	
	Ebn at Noon	760	796	791	769	776	764	771	761	737	702	699	723	
	Edh at Noon	81	109	142	176	174	172	160	148	134	110	81	70	
All-Sky Solar Radiation	RadAvg	1.56	2.47	3.73	4.48	5.03	5.88	5.92	5.09	3.91	2.49	1.52	1.30	
	RadStd	0.20	0.31	0.52	0.65	0.47	0.46	0.46	0.40	0.34	0.33	0.24	0.17	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (114 neighbors)	N/A	N/A	N/A	+0.73	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	+0.72	+0.40	+0.31	N/A	N/A	+59	+16

Nomenclature: See separate page